Detecting Heart Abnormality Using ECG with CART and Ensemble Models

UE24CS352A — Machine Learning Mini-Project

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1. Problem Statement

Heart disease is one reason to study automated ECG analysis. This project uses the labelled UCI Arrhythmia dataset to classify ECG records into the diagnostic classes provided with the data. We use Classification and Regression Trees (CART) as an interpretable baseline and compare it with ensemble methods and class-balanced Logistic Regression. The aim is to see how these models perform on the same test set and how class imbalance affects the results.

2. Dataset

The UCI Arrhythmia dataset contains 452 records and 279 input attributes before feature removal. The attributes include patient measurements and ECG signal/channel features. Missing values are marked with "?" in the raw file. Removing feature 14 leaves 278 input features. Thirteen target labels appear in this dataset: 1–10, 14, 15 and 16; there are no records for classes 11–13. The class distribution is highly imbalanced: class 1 has 245 records, while class 8 has only 2.

3. Approach

The script treats "?" as missing and removes feature 14, following the reference work. It then makes a stratified 70/30 train-test split with random_state=42, giving 316 training records and 136 test records. Median imputation is part of a Scikit-learn Pipeline: the medians are learned from the training data and reused for the test set. Standard scaling is used only for Logistic Regression. All 13 observed classes are kept.

We evaluate six pipelines: CART with Gini impurity, class-balanced CART, Random Forest, Extra Trees, Gradient Boosting and class-balanced Logistic Regression. Random Forest and Extra Trees combine predictions from multiple trees. Gradient Boosting adds trees one after another, with each stage aiming to correct errors from earlier stages. For the balanced models, class weights give less frequent classes more influence during training.

4. Evaluation

All six models are evaluated on the same holdout test set using accuracy, balanced accuracy, Macro F1 and Weighted F1. Macro F1 gives each observed class equal weight, so it helps show how the models handle an imbalanced dataset. We also calculate multiclass ROC-AUC using a One-vs-Rest (OvR) approach, reported as macro and support-weighted averages. For the model with the highest Macro F1, the script saves a per-class classification report and a labelled confusion matrix. Because we select the winner on this same holdout set, its reported score may be optimistic. A stronger follow-up would select the model using training-only validation and evaluate it on a separate, untouched test set.

5. Implementation

src/train_compare_ecg_models.py loads or downloads the dataset, builds the six pipelines, trains and compares them, then selects the model with the highest holdout Macro F1. It saves the fitted pipeline to results/best_model.joblib and writes comparison metrics, ROC-AUC scores and plot, predictions, classification reports, confusion matrix and run summaries to results/. src/predictor.py loads the saved pipeline and predicts the class for a selected zero-based row in arrhythmia.data. The README explains how to install the dependencies and run both scripts.

6. Results and Reference Comparison

On the stratified 70/30 split, Gradient Boosting has the highest Macro F1. The table compares all six models on the same 136-record test set. Accuracy and balanced accuracy are shown as percentages; F1 and ROC-AUC scores use a 0–1 scale.

Model	Accuracy	Balanced accuracy	Macro F1	Weighted F1	Macro AUC (OvR)	Weighted AUC (OvR)
Gradient Boosting	**73.53%**	**46.66%**	**0.4036**	**0.6985**	**0.8091**	**0.8708**
CART (Gini)	69.12%	36.98%	0.3657	0.6780	0.6674	0.7733
CART (balanced)	59.56%	36.23%	0.3457	0.5930	0.6592	0.7155
Logistic Regression (balanced)	63.24%	39.08%	0.3433	0.6143	0.7332	0.8130
Random Forest	65.44%	30.88%	0.2990	0.5523	0.8775	0.8890
Extra Trees	62.50%	28.52%	0.2765	0.5124	0.8564	0.8754

Compared with standard CART on this split, Gradient Boosting raises accuracy from 69.12% to 73.53% (+4.41 percentage points) and balanced accuracy from 36.98% to 46.66% (+9.68 points). Macro F1 rises from 0.3657 to 0.4036, and Weighted F1 rises from 0.6780 to 0.6985. This is an improvement over our CART baseline on this split, but it does not guarantee better performance on other data.

Random Forest has the highest macro OvR AUC (0.8775) and weighted OvR AUC (0.8890). Extra Trees is next, at 0.8564 and 0.8754. Gradient Boosting still has the highest Macro F1. These rankings can differ because ROC-AUC measures how well a model ranks class scores across thresholds, while F1 is based on the final predicted classes.

In “Detecting Heart Abnormality using ECG with CART,” Rajbhandary, Zhou and Garcia Jr. report 68.46% CART-Gini accuracy for a 70/30 split under the V14-imputed setting (Table 3). Our Gradient Boosting accuracy of 73.53% is numerically higher, but this is not a direct, controlled comparison because the preprocessing and evaluation details may differ. The paper’s approximate 80% figure refers to its broader reported result, not that specific holdout setting.

Gradient Boosting has zero recall for classes 5, 7, 8, 14 and 16. The test set contains only one record each for classes 7, 8 and 14, so those class-level scores can change sharply with a single prediction. Macro F1 and balanced accuracy help show these minority-class weaknesses, which overall accuracy alone can hide.

7. Conclusions

On this test split, Gradient Boosting gives the best Macro F1 (0.4036) and accuracy (73.53%), while Random Forest gives the best ROC-AUC (macro 0.8775; weighted 0.8890). These results come from one split and should be treated as preliminary. The dataset is strongly imbalanced, some classes have very few examples, and selecting the winner on the same holdout set can make its score optimistic. A stronger evaluation would select the model using training-only validation and reserve an untouched test set for final scoring. This project is an academic experiment, not a clinical diagnostic system.

8. References

1. UCI Machine Learning Repository, “Arrhythmia” dataset. DOI: 10.24432/C5BS32.

2. Paurakh Rajbhandary, Benjamin Zhou and Gaspar Garcia Jr., “Detecting Heart Abnormality using ECG with CART,” Table 3, overall accuracy results.